

Inflammatory Fat and Lipolysis

Lipolysis and lipid droplet (PLIN1), desaturation (SCD, FADS1), fatty acid activation (ACSL1), fat distribution (LYPLAL1, GRB14, RSPO3, KLF14, PPARG, ADIPOQ, FTO), liver (PNPLA3, TM6SF2, MBOAT7, HSD17B13, GCKR, PiMZ), incretin and glucagon receptors (GLP1R, GIPR, GCGR), visceral fat scores, gluteo-femoral and hepatic

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The gluteal-femoral fat, the "safe" fat, has a high genetic score (percentile 95.3), and the visceral fat adjusted to total adiposity is at percentile 43.3 (typical). Genetic **FABP4** (a pro-inflammatory adipokine) is at percentile 18.5 and adiponectin (anti-inflammatory) has a loci load at 88.9. Sérgio F does not have the central distribution alleles of **LYPLAL1**, **GRB14/COBLL1** or **RSPO3**. In the liver, it does not have **TM6SF2 E167K** and has 1 allele that protects HSD17B13. **GIP** and glucagon receptors are of the common type (G/G).

Unfavourable. The percentage of fat in the genetic trunk is at percentile 98.4 and the subcutaneous abdominal fat adjusted in the 87.2. It has 1 risk allele in KLF14 (larger adipocytes, plus visceral fat) and 1 in FTO. In the liver, the known findings are repeated: PNPLA3 I148M heterozygous, **PiMZ** (SERPINA1 Z), **MBOAT7** and **GCKR** with 1 allele; the load of fat loci in the liver is at percentile 79.3.

Lipolysis and desaturation. **PLIN1 11482A** is homozygous (T/T: less perilipin and more basal lipolysis, associated in some studies with more loss of fat with caloric deficit). **SCD** does not have alleles that increase the monounsaturated (loci load at 66.2) and **FADS1** is T/T (complete activity). **ACSL1** is not GG.

What does it mean? The genetic pattern is that of central fat in the trunk with good peripheral storage capacity: the trunk accumulates, but the gluteal-femoral fat protects and the visceral adjusted is typical. With retatrutide, genes do not predict a bad response: GLP1R 168 is Ser/Being (associated in some studies with the best response), 7 is Leu/Leu (worst glycemic response in small studies) and GIPR and GCGR are common. The most important point remains the liver (PNPLA3 + PiMZ), already treated in §51.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 22 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 18 of the chip, 4 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 6 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 10 PGS Catalog scores calculated locally with 4,601,629 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 510^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 4593 loci, 4103 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In the visceral, abdominal, trunk, FABP4 and fatty liver scores, a higher percentile = more fat or more risk. In gluteal-femoral fat, a higher percentile = better (protective fat). 'Adjusted' : adjusted to total adiposity (measures distribution and not quantity).

Visceral and central fat

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Forecast visceral fat	PGS000844 (Aly 2021, 518 variants)	511 518 (weighted coverage 99%)	73.3	TYPICAL	typical
Visceral fat (adjusted to total adiposity)	PGS002813 (Agrawal 2022, LDpred2, 1.13 million variants)	1,088,675 1,125,301 (weighted coverage 95%)	43.3	TYPICAL	typical
Subcutaneous abdominal fat	PGS002811 (Agrawal 2022, LDpred2, 1.13 million variants)	1,088,675 1,125,301 (weighted coverage 96%)	87.2	HIGH	unfavourable
Waist-hip ratio (adjusted to total adiposity)	PGS003485 (Zhang 2022, LDpred2, 785 thousand variants)	770,261 785,126 (weighted coverage 97%)	52.1	TYPICAL	typical
Percent of trunk fat	PGS001105 (Tanigawa 2022, 26 thousand variants)	22,975 25,649 (weighted coverage 91%)	98.4	HIGH	unfavourable

Protective fat

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Gluteal-femoral fat (adjusted)	PGS002812 (Agrawal 2022, LDpred2, 1.13 million variants)	1,088,675 1,125,301 (weighted coverage 96%)	95.3	HIGH	favourable

Adipocyte

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
FABP4 (adipocyte fatty acid binding protein)	PGS000237 (Folkersen 2020, LDpred, 552 thousand variants)	541,771 552,343 (weighted coverage 98%)	18.5	LOW	favourable

Fatty liver

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Liver fat (MRI)	PGS002245 (Haas 2021, 8 variants)	8 8 (weighted coverage 100%)	65.4	TYPICAL	typical
Non-alcoholic fatty liver	PGS002282 (Schnurr 2022, 68 variants)	68 68 (weighted coverage 100%)	23.9	TYPICAL	typical
Non-alcoholic fatty liver	PGS012549 (Tang 2023, 10 variants)	10 10 (weighted coverage 100%)	50.6	TYPICAL	typical

4. Key variants by topic

Variants of lipolysis, desaturation, distribution of fat, liver and incretin receptors.

Lipolysis, desaturation and activation of fatty acids

Lipolysis releases fatty acids from the stored triglycerides: perilipina-1 (PLIN1) protects the droplet, ATGL and HSL cuts the triglycerides. SCD1 transforms saturated fat into monounsaturated and ACSL1 activates fatty acids. When the adipocyte overflows, saturated fatty acids and ceramides activate TLR4 and inflammasome: it is the "proinflammatory fat".

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PLIN1	rs894160	11482G>A	T/T	chip	T (2 copy/s)	without judgment	Perilipina-1 (11482G>A; A = T in strand +): allele A is associated with less perilipin in the droplet, more basal lipolysis and, in some studies, more loss of fat with caloric restriction	Weak
SCD	rs3793770		G/G	imputation DR2=1.00	T (0 copy/s)	without judgment	Stearoyl-CoA desaturase (SCD1): converts saturated fat (16:0, 18:0) into monounsaturated fat (16:1, 18:1); the effect allele increases the proportion of monounsaturates (plus SCD activity) [GWAS Catalog: Monounsaturated fatty acids to total fatty acids percentile, T allele, p=1E-13, PMID 40545721]	Moderate
SCD (OLMALINC)	rs56314467		C/C	imputation DR2=0.88	G (0 copy/s)	without judgment	Rare SCD regulator (lncRNA OLMALINC): proportion of monounsaturated [GWAS Catalog: Monounsaturated fatty acids to total fatty acids percentile, allele G, p=2E-24, PMID 40545721]	Weak
FADS1	rs174547		T/T	chip	C (0 copy/s)	without judgment	Delta-5 desaturase: C allele reduces activity (less arachidonic acid and endogenous EPA, plus linoleic acid); with fish oil, EPA/DHA comes directly from the diet	Strong
ACSL1	rs9997745		A/A	chip	G (0 copy/s)	neutral	Acyl-CoA synthetase 1 (active fatty acids for oxidation or storage): the genotype GG was associated with more metabolic syndrome, especially with a high saturated fat diet (Phillips 2010; limited replication)	Weak

Fat and adipose tissue distribution

Visceral fat and ectopic fat are inflammatory (IL-6, TNF, less adiponectin). The gluteal-femoral fat, on the contrary, stores fatty acids safely and protects metabolism.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
LYPLAL1	rs2605100		A/A	chip	G (0 copy/s)	favourable	Lysophospholipase-like 1: central fat distribution (especially in women)	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GRB14 / COBLL1	rs10195252		C/C	chip	T (0 copy/s)	favourable	Locus GRB14/COBLL1: central fat, triglycerides and insulin resistance (light lipodystrophy profile) [GWAS Catalog: Waist-to-hip ratio adjusted for adiposity, T allele, beta 0.031 [0.027-0.035] unit increase, p=9E-69, PMID 30575882]	Moderate
RSPO3	rs9491696		C/C	chip	G (0 copy/s)	favourable	R-spondine 3 (via WNT): expansion of gluteal-femoral vs abdominal adipose tissue [GWAS Catalog: Waist-hip ratio, G allele, beta 0.040 unit increase, p=2E-40, PMID 20935629]	Moderate
KLF14	rs4731702		C/T	chip	C (1 copy/s)	unfavourable (light)	Maternal expression regulator in adipose tissue: C allele is associated with larger adipocytes, more visceral fat and 2 type diabetes (Small 2018)	Moderate
PPARG	rs1801282	Pro12Ala	C/C	chip	G (0 copy/s)	neutral	Pro12Ala: G allele (Ala) improves insulin sensitivity and reduces 2 type diabetes (~15-20% per allele)	Strong
ADIPOQ	rs17300539	-11391 G>A	G/G	chip	A (0 copy/s)	neutral	-11391 G>A promoter: allele A increases adiponectin (anti-inflammatory hormone of the adipocyte)	Moderate
FTO	rs9939609		T/A	chip	A (1 copy/s)	unfavourable (light)	Common FTO variant (via IRX3/IRX5, less adipocyte thermogenesis): more adiposity and appetite	Strong

Fatty liver (genetic)

The same liver fat genes from the Liver report (\$51), here seen on the lipolysis side: PNPLA3 148M stops the output of triglycerides from the hepatic droplets.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PNPLA3	rs738409	I148M	C/G	chip	G (1 copy/s)	unfavourable (light)	Ile148Met: the mutant protein accumulates in the droplets and stops liver lipolysis; the largest fatty liver genetic factor (details in the Liver report)	Strong
TM6SF2	rs58542926	E167K	C/C	chip	T (0 copy/s)	favourable	Glu167Lys: less exportation of VLDL, more fat in the liver (but lower LDL)	Strong
MBOAT7	rs641738		C/T	chip	T (1 copy/s)	unfavourable (light)	MBOAT7 variant: less remodeling of phosphatidylinositol, more fat and liver fibrosis	Moderate
HSD17B13	rs6834314		A/G	chip	G (1 copy/s)	favourable	HSD17B13 protective splicing allele marker (rs72613567): less inflammation and hepatic fibrosis	Moderate
GCKR	rs1260326	P446L	T/C	chip	T (1 copy/s)	unfavourable (light)	Pro446Leu: more hepatic lipogenesis and triglycerides (but lower fasting glucose)	Strong
SERPINA1	rs28929474	PiZ	C/T	chip	T (1 copy/s)	unfavourable	PiZ (Glu342Lys): alpha-1 antitrypsin Z polymerizes in the reticulum of hepatocytes; in PiMZ, more risk of fatty liver progress to fibrosis (see reports Liver and Lung)	Strong

Retatrutide receptors (GIP, GLP-1, glucagon)

Retatrutide is a triple agonist of GIP, GLP-1 and glucagon receptors. The variants of these receptors can, in theory, change the response; the evidence in humans is small and does not exist for retatrutide in particular.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GLP1R	rs6923761	Gly168Ser	A/A	imputation DR2=1.00	A (2 copy/s)	without judgment	Gly168Ser: in some small studies, allele A was associated with more loss of fat and appetite with GLP-1 agonists (liraglutide, exenatide)	Weak
GLP1R	rs10305420	Pro7Leu	T/T	imputation DR2=0.98	T (2 copy/s)	without judgment	Pro7Leu: T allele was associated with lower glycemic response to GLP-1 agonists in small studies	Weak
GIPR	rs1800437	Glu354Gln	G/G	chip	C (0 copy/s)	without judgment	Glu354Gln: C allele (Gln) desensitizes the GIP receptor; less adiposity and more bone density	Moderate
GCGR	rs1801483	Gly40Ser	G/G	chip	A (0 copy/s)	without judgment	Gly40Ser: Allele A reduces the response to glucagon (relevant to the glucagon component of retatrutide)	Weak

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of visceral/abdominal fat, waist-hip ratio adjusted, proportion of monounsaturated (without judgment), adiponectin and fat in the liver, with alleles oriented towards the highest value, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Visceral / abdominal fat	356 (325)	328 of 650	325	70.6	typical
Waist-hip ratio (adjusted)	2628 (2425)	2408 of 4850	2421	44.5	typical
Monounsaturated (SCD activity)	1493 (1246)	608 of 2492	1245	66.2	without judgment
Adiponectin	83 (74)	84 of 148	74	88.9	favourable
Fat in the liver	33 (33)	29 of 66	33	79.3	typical

6. Favourable

- **High gluteal-femoral fat** (percentile 95.3): more ability to store fat outside the abdomen and liver; protects insulin sensitivity. **Moderate.**
- **Typically adjusted visceral fat** (percentile 43.3; visceral/abdominal fat loci load at 70.6); waist-to-hip ratio adjusted in 52.1 (loci at 44.5). **Moderate.**
- **Low genetic FABP4** (percentile 18.5): less of this pro-inflammatory and atherogenic adipokine. **Weak moderate.**
- **High genetic adiponectin** (Loci load at percentile 88.9): more of this anti-inflammatory hormone and insulin sensitizer. **Moderate.**
- **LYPLAL1, GRB14/COBLL1 and RSPO3 without central fat alleles. Moderate.**
- **TM6SF2 C/C** (without E167K) and **HSD17B13 A/G** (1 allele protective against inflammation and liver fibrosis). **Strong/Moderate.**
- **Polygenic fatty liver** (23.9 and 50.6 percentiles). **Moderate.**

7. Unfavourable

- **High genetic trunk fat:** percentage of trunk fat at percentile 98.4; subcutaneous fat adjusted at 87.2; visceral fat predicted at 73.3. **Moderate.**
- **KLF14 rs4731702 C/T** and **FTO rs9939609 T/A** (1 risk allele each): larger adipocytes and more adiposity. **Moderate/Strength.**
- **PNPLA3 I148M C/G** + **PiMZ** (SERPINA1 C/T): the two genetic cofactors of the liver (detail in §51 and in the Pulmonary report). **Strong.**
- **MBOAT7 C/T** and **GCKR P446L T/C** (1 allele each): more hepatic fat and more lipogenesis. Liver fat loci load at percentile 79.3; liver fat (8 variants) in the 65.4. **Moderate.**
- **PPARG Pro/Pro** and **ADIPOQ G/G**: without protective alleles (common genotypes). **Neutral.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Dysfunction of adipose tissue	High trunk fat (percentile 98.4), but protective gluteum-femoral (percentile 95.3); KLF14 with 1 allele	Mixed
Insulin resistance	GRB14/COBLL1 and LYPLAL1 without risk alleles; PPARG without Ala12	Neutral
Chronic Inflammation	Low genetic FABP4 and high adiponectin (loci at percentile 88.9); high trunk fat	Mixed
Metabolic Homeostasis Decline	PNPLA3 148M, MBOAT7 and GCKR with 1 allele each (ectopic fat in the liver)	Unfavourable (mild)
Autophagy / Deregulated Macroautophagy	PNPLA3 148M hinders lipophagy (autophagy of droplets) in the hepatocyte	Unfavourable (mild)
Protein Aggregation	PiMZ: alpha-1 antitrypsin Z polymers in the hepatocyte reticulum	Unfavourable
Endoplasmic Reticulum Disorder	PiZ and excess of saturated fatty acids cause reticulum stress	Unfavourable (mild)
Thermogenesis decline	FTO with 1 allele (less adipocyte thermogenesis)	Unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **During retatrutide, protect lean mass:** high protein distributed by all meals and strength training 2–4 times a week; trunk fat and visceral fat also respond to aerobic training. *Suggestion (not applied).*
2. **Dietary fat quality:** exchange saturated fat (fat, butter, coconut oil and palm oil) for extra virgin olive oil, dried fruits and fish; less substrate for SCD and less saturated fatty acids in the adipocyte and liver. *Suggestion (not applied).*
3. **Fiber at most and free sugars and fructose at least** (sumes, soft drinks, sweets): fructose feeds the hepatic lipogenesis, which GCKR and PNPLA3 amplify. Zero alcohol (already fulfills). *Suggestion (not applied).*
4. **Omega-3 3.2–4 g/day (already in the plan):** low triglycerides and moderate hepatic fat. *Suggestion (not applied).*
5. **Liver (PNPLA3 148M + PiMZ):** follow the Liver report plan; follow the evolution of liver fat with retatrutide and consider a hepatic elastography if the fat does not decrease. *Suggestion (not applied).*
6. **Do not adjust the retatrutide by genes:** the variants GLP1R, GIPR and GCGR have weak evidence and no study with retatrutide; the real response (body composition, liver, glucose) is the guide. *Suggestion (not applied).*
7. **Rhythm of meals:** meals concentrated in the first half of the day and regular sleep improve nocturnal lipolysis and insulin sensitivity of the adipocyte. *Suggestion (not applied).*

10. Limitations

- The "adjusted" fat scores measure the distribution and not the amount; the actual fat depends mainly on the diet, exercise and treatment in progress.
- The effects of PLIN1, ACSL1 and incretin receptors come from small studies, with limited replication; none was studied with retatrutide.
- The proportion of monounsaturates in the blood mixes SCD activity and diet; loci are shown to be unbiased.
- The chip does not read the HSD17B13 rs72613567 (insertion); the rs6834314 marker is used.
- The load of SNPs in linkage disequilibrium is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (10 scores: visceral, abdominal, trunk and gluteal-femoral fat, waist-to-hip ratio, FABP4, liver fat and fatty liver). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, empirical calibration in the same reference population.
- **Interpretation:** reference literature (Smith 2015 and Corella 2006 for PLIN1; Phillips 2010 for ACSL1; Small 2018 for KLF14; Agrawal 2022 for fat distribution Romeo 2008 for PNPLA3; Abul-Husn 2018 for HSD17B13; Javorský 2016 and Luis 2014 for GLP1R), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Lipolysis, desaturation and activation of fatty acids	PLIN1	rs894160	T/T	chip	T	2	strand +	2 copy of the effect allele
Lipolysis, desaturation and activation of fatty acids	SCD	rs3793770	G/G	imputation DR2=1.00	T	0	strand +	0 copy of the effect allele
Lipolysis, desaturation and activation of fatty acids	SCD (OLMALINC)	rs56314467	C/C	imputation DR2=0.88	G	0	strand + (palindromic; assumed +)	0 copy of the effect allele
Lipolysis, desaturation and activation of fatty acids	FADS1	rs174547	T/T	chip	C	0	strand +	0 copy of the effect allele
Lipolysis, desaturation and activation of fatty acids	ACSL1	rs9997745	A/A	chip	G	0	strand +	AA
Fat and adipose tissue distribution	LYPLAL1	rs2605100	A/A	chip	G	0	strand +	without the risk allele
Fat and adipose tissue distribution	GRB14 / COBLL1	rs10195252	C/C	chip	T	0	strand +	without the risk allele
Fat and adipose tissue distribution	RSPO3	rs9491696	C/C	chip	G	0	strand + (palindromic; assumed +)	without the risk allele
Fat and adipose tissue distribution	KLF14	rs4731702	C/T	chip	C	1	strand +	1 copy of risk allele
Fat and adipose tissue distribution	PPARG	rs1801282	C/C	chip	G	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
Fat and adipose tissue distribution	ADIPOQ	rs17300539	G/G	chip	A	0	strand +	without the protective allele (common genotype)
Fat and adipose tissue distribution	FTO	rs9939609	T/A	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
Fatty liver (genetic)	PNPLA3	rs738409	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Fatty liver (genetic)	TM6SF2	rs58542926	C/C	chip	T	0	strand +	without the risk allele
Fatty liver (genetic)	MBOAT7	rs641738	C/T	chip	T	1	strand +	1 copy of risk allele
Fatty liver (genetic)	HSD17B13	rs6834314	A/G	chip	G	1	strand +	1 copy of the protective allele
Fatty liver (genetic)	GCKR	rs1260326	T/C	chip	T	1	strand +	1 copy of risk allele
Fatty liver (genetic)	SERPINA1	rs28929474	C/T	chip	T	1	strand +	PIMZ (Z carrier)
Retatrutide receptors (GIP, GLP-1, glucagon)	GLP1R	rs6923761	A/A	imputation DR2=1.00	A	2	strand +	2 copy of the effect allele
Retatrutide receptors (GIP, GLP-1, glucagon)	GLP1R	rs10305420	T/T	imputation DR2=0.98	T	2	strand +	2 copy of the effect allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Retatrutide receptors (GIP, GLP-1, glucagon)	GIPR	rs1800437	G/G	chip	C	0	strand + (palindromic; assumed +)	0 copy of the effect allele
Retatrutide receptors (GIP, GLP-1, glucagon)	GCGR	rs1801483	G/G	chip	A	0	strand +	0 copy of the effect allele